

Week 5

Literature Review

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2026-09-18

Today's Icebreaker

- ▶ If you've met with your potential advisor(s), how has your original idea changed?
 - ▶ If you haven't met with them, what field of research are you thinking of looking at to match research from your potential advisor(s)?

JP's Response

- ▶ Additional focal points
 - ▶ No longer focused on intervention design and now focused on prediction
 - ▶ Thinking through pros and cons of interviews vs focus groups

Announcement

- ▶ Would you rather have November 17th be the first round of final assignment presentations?
 - ▶ Thanksgiving week would be a check in + work on final written project
 - ▶ **All** final presentations would need to be submitted by **November 17th**
- ▶ Keep presentations for November 24th and have a lottery system for who gets to present on which day
 - ▶ You would have to be here for both presentation days because part of the assignment is giving feedback

Keywords

- ▶ What are some keywords that come to mind when thinking about your research topic?
- ▶ Keywords include:
 - ▶ Math*
 - ▶ (“Self-efficacy” OR “self efficacy”)
 - ▶ Grade*
 - ▶ College OR university OR higher education
 - ▶ Cluster*
 - ▶ unsupervised
 - ▶ ((latent AND “mixture modeling”) OR class OR profile)
 - ▶ Specific clustering algorithms

How to Look for Articles

- ▶ Use keywords
- ▶ Don't type out full statements
- ▶ Use forward & backward searches
- ▶ AI Options*

AI Use

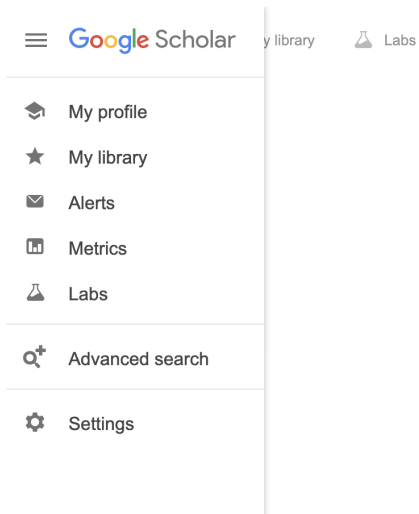


Photo by Aleksandar Cvetanovic

How to Look for Articles

- ▶ Google Scholar
- ▶ Can use the [Berkeley Library website](#)

Connect Google Scholar to UCB Library



Connect Google Scholar to UCB Library



 Settings

Search results

Languages

Library links

Account

Browser extensions

Show library access links for (choose up to five libraries):



e.g., Harvard

☒ UNIVERSITY OF CALIFORNIA BERKELEY - ProQuest Fulltext

☒ University of California Berkeley - Get it at UC

☒ Open WorldCat - Library Search

Online access to library subscriptions is usually restricted to patrons of that library. You may need to login with your library password, use a campus computer, or configure your browser to use a library proxy. Please visit your library's website or ask a local librarian for assistance.

Save

Cancel

To retain settings, you must turn on [cookies](#)

Keywords

- ▶ Use AND, OR, (), -, "", ~, *
- ▶ Artificial Intelligence OR AI
- ▶ Artificial Intelligence AND Learning
- ▶ (Artificial Intelligence OR AI) AND Learning
- ▶ (Artificial Intelligence OR AI) AND Learning AND -Deep (- = don't include)
- ▶ ("Artificial Intelligence" OR AI) AND Learning AND -"Deep Learning" (" " = Look up exact term)
- ▶ ("Artificial Intelligence" OR AI) AND (Learning OR ~education) AND -"Deep Learning" (~ = Look up synonyms)
- ▶ ("Artificial Intelligence" OR AI) AND ((Learning OR education) AND (college OR university OR "higher education")) AND -"Deep Learning" (= Any word with the same beginning)

Keywords

- ▶ Sometimes your searches can sound problematic
 - ▶ Example: (Latina OR Latino OR Latinx OR Latine OR Hispanic OR *Mexican*)
 - ▶ Why not Latin*?
- ▶ [Resource here](#)

Forward Searches

- ▶ You find an amazing article
 - ▶ Maybe it is too old
 - ▶ Has useful information
- ▶ Look at who cites that paper
 - ▶ Re-use your keywords in this new search

Example of Forward Search

("Artificial Intelligence" OR AI) AND ((Learning OR educat*) AND (college OR



About 7,660,000 results (0.15 sec)

Artificial intelligence in higher education: The state of the field

[H. Crompton](#), [D. Burke](#) - ... of educational technology in **higher education**, 2023 - Springer

... The final code in this study highlighted the use of **AI** in managing student **learning**, including **learning** analytics, curriculum sequencing, instructional design, and clustering of students. ...

☆ Save [W](#) Cite Cited by 4101 [Related articles](#) [All 16 versions](#) [Web of Science: 870](#) [36](#)

[PDF] [springer.com](#)

✦ Quick read - [odu.edu](#)

Full View

Revolutionizing education: Artificial intelligence empowered learning in higher education

[H.U. Rahman](#), [R. Koodikal](#) - Cogent Education, 2024 - Taylor & Francis

... Since the objective of this research is to understand the impact of **AI** on **higher education** and investigate the adoption and implementation of **AI** in **higher education**, we developed a ...

☆ Save [W](#) Cite Cited by 479 [Related articles](#) [All 5 versions](#) [36](#)

[PDF] [tandfonline.com](#)

✦ Quick read

Full View

Example of Forward Search



About 4,101 results (0.04 sec)

Artificial intelligence in higher education: The state of the field

☐ Search within citing articles

[HTML] Artificial intelligence in teaching and teacher professional development: A systematic review

X Tian, G Zhang, M Li Jing - Computers and education: Artificial Intelligence, 2025 - Elsevier

Abstract: The application of Artificial Intelligence (AI) technology in education is increasingly recognized as a key driver of educational innovation. While extensive literature exists on the ...

☆ Save  Cite Cited by 934 Related articles All 3 versions 

[HTML] sciencedirect.com

✚ Quick read

AI-driven adaptive learning for sustainable educational transformation

Y Sitnikova, V Grebennikova - Sustainable ..., 2025 - Wiley Online Library

This paper scrutinizes how adaptive learning technologies and artificial intelligence (AI) are transforming today's education by making it personalized, accessible, and efficient as well. ...

☆ Save  Cite Cited by 1096 Related articles All 8 versions Web of Science: 245 

[PDF] wiley.com

✚ Quick read

Full View

Role of AI chatbots in education: systematic literature review

L Lalasize, M Sijodha, L Machadze - International Journal of Educational ..., 2023 - Springer

AI chatbots shook the world not long ago with their potential to revolutionize education systems in a myriad of ways. AI chatbots can provide immediate support by answering ...

☆ Save  Cite Cited by 1990 Related articles All 8 versions Web of Science: 506 

[PDF] springer.com

Full View

Example of Forward Search



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☒ [Search within citing articles](#)

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[PDF] wiley.com

◆ Quick read

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☆ Save ⚙ Cite Cited by 1096 Related articles All 8 versions Web of Science: 245 ⓘ

[PDF] wiley.com

✚ Quick read

Full View

Example of Forward Search

- ▶ Keep going
- ▶ Add filters on the left sidebar
- ▶ read abstracts to see what is useful
- ▶ Diversify authors
 - ▶ Don't just cite one group of authors
 - ▶ Looks like you didn't read enough literature

Backward Searches

- ▶ While reading a useful article
- ▶ You find an interesting claim
 - ▶ Background information
 - ▶ Finding similar to the conclusion made in the current article
 - ▶ Check the references
- ▶ Look at the reference in Google Scholar
 - ▶ Can then forward search from this article

AI Options

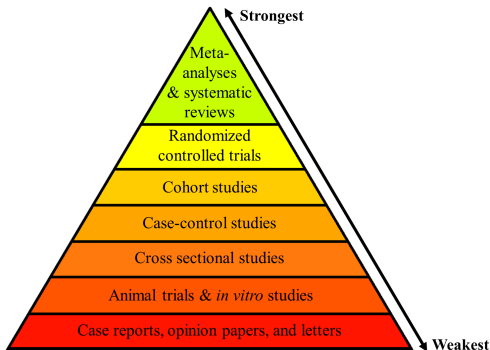
- ▶ Consensus
- ▶ Elicit
- ▶ Make sure to double check references

Rules

- ▶ Seminal papers can be from any time
 - ▶ Usually someone has written about seminal papers to update them
- ▶ Empirical papers
 - ▶ Papers that run some form of analysis
 - ▶ Past ~10 years
 - ▶ Field dependent
- ▶ Understand the hierarchy of evidence when gathering literature

Hierarchy of Evidence

Hierarchy of Scientific Evidence



Types of References to Use

- ▶ Non-academic articles
 - ▶ Online research briefs (e.g., Pew Research Center)
 - ▶ Mainly used in hook of introduction
- ▶ Peer-reviewed articles/conference proceedings
 - ▶ Empirical research
 - ▶ Systematic Reviews/Scoping Reviews/Meta-Analyses

Find Research Topic

- ▶ Central idea for your research
- ▶ Should be researched if
 - ▶ There is a gap in the literature
 - ▶ You have access to data that would answer the topic
 - ▶ Advances your academic career

Point of the Lit Review

- ▶ Provides a framework showing importance of this research
- ▶ You build your argument with supporting or contradicting findings
- ▶ Helps solidify your research topic when identifying the gap(s) in the literature

Literature Maps

- ▶ Map out how your literature will support your argument
 - ▶ Create categories to organize your paper into
 - ▶ Google [NotebookLM](#)
- ▶ Thinking like this can help with the writing process (when we get there)

How to read articles

- ▶ If using AI, go ahead and use that for your initial search
- ▶ Read **abstracts only** the first time through
- ▶ Save articles you think may be useful
 - ▶ Zotero
- ▶ Then you read the articles

Example Article

- ▶ Incorporating Deep Learning Into System Dynamics
- ▶ Use your own article if you want

Abstracts

- ▶ Is this relevant?
 - ▶ No → move on
 - ▶ Yes → save to Zotero

Introduction

- ▶ New topic = read
 - ▶ Citations may be helpful for developing research idea
 - ▶ Know about the topic → ignore/skim
 - ▶ Look at the final paragraph in this section → RQs/hypotheses/aims/purpose/etc.
- ▶ Contains theory

Methods

- ▶ Very important (expect to add a lot here in annotated bibliographies)
- ▶ Contains protocol for data collection
- ▶ Cleaning methods
- ▶ Description of data
- ▶ Analytic Technique
- ▶ What the researchers did

Results

- ▶ Main findings of the study
 - ▶ Dense ()
- ▶ I usually read it to get specifics
 - ▶ Actual estimation values

Discussion/Conclusion

- ▶ Reports findings in clear terms
 - ▶ If something is interesting here THEN go to the results to understand more
- ▶ I write out main findings of the study in my annotated bibliographies

Future Studies/Limitations

- ▶ Can find inspiration of how you might be able to expand on research that has been completed
- ▶ You can also take some of the future direction(s) and apply it to your work
 - ▶ If relevant

If you haven't already, get an article

Annotated Bibliographies

- ▶ Different philosophies of how to create annotated bibliographies
 - ▶ Short paragraphs with all information (~150 words)
 - ▶ Feels like another abstract
- ▶ Spreadsheets

Example

- ▶ Make a copy or you could make something similar
 - ▶ [Link here for example spreadsheet](#)
- ▶ If you would like to use a document instead
 - ▶ [Link here for example document](#)

Annotated Bibliographies Assignment

- ▶ Due Oct 4th at 11:59pm
- ▶ ~10 articles
 - ▶ No lower or upper cap
 - ▶ This is for you; you decide how many articles you need